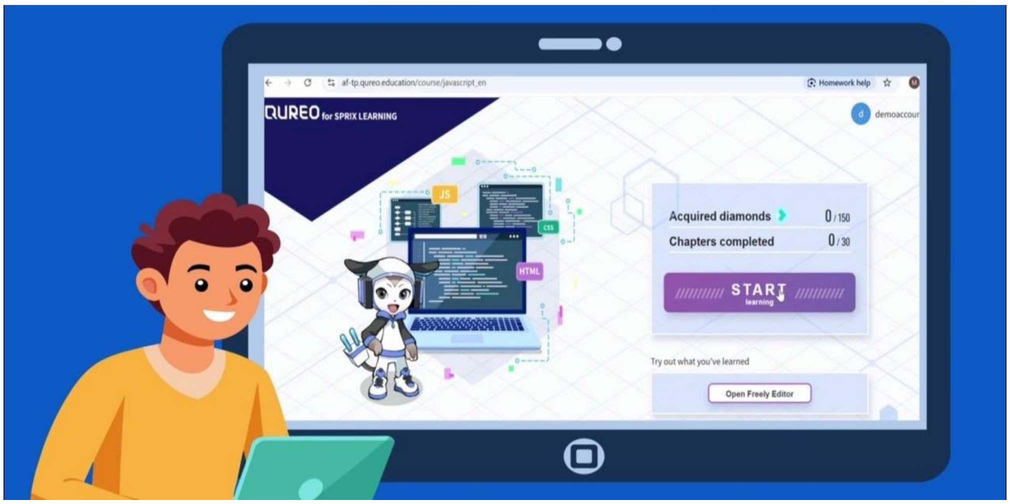
PONY

Programming and AI



Your guide to

JavaScript

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Chapter 1

Programming

It means instructions provided to a computer , these instructions are called a "program"

programming allows you to create:

1. games
2. websites
3. applications

- There are more than 100 programming languages in the world

HTML	CSS	JavaScript
Build the framework	Create appearance	Create movement
HTML	HTML + CSS	HTML + CSS JavaScript

- The black screen where you write your program is called:
  text editor or console.

Example:

- Write a program to output “Hello”.

JavaScript

```
console.log("Hello");
```

NOTES

- Text should be added between " "
- To write a comment: start the sentence with //

Review

1. **Select the correct program to output "Hello" to the console.**
    a) log.console("Hello");
    b) console.log("Hello");
    c) consolelog("Hello")
    d) console.log(Hello);

2. **Create a program to output "Good morning" to the console.**

....

3. **Select the correct way to write your comment.**
    a) /Write your comments here./
    b) //Write your comments here.
    c) /Write your comments here.
    d) Write your comments here.//

Chapter 2
Calculations and Strings

1 To output *text*:

write text with " " in *console.log( );*

2 To output *calculation* results:

write the calculation formula in brackets without " " in *console.log( );*

JavaScript | Run Code

```
console.log("Hello World");
console.log("2+2");
console.log(2+2);
```

Console

```
Hello World
2+2
4
```

- Add: console.log(32 +48)
- Subtract: console.log(100 - 40)
- Multiply: console.log(5 * 9)
- Divid: console.log(10/2)

To connect characters, numbers and calculations in a one-line program, you can use "+" to join each part:

JavaScript

```
console.log("Set of 2 coffes "+(250*2)+"coins");
```

JavaScript

```
console.log("100 + 200" + “is” + (100 + 200));
```

Result console

```
100 + 200  is  300
```

Revision

1. **Select the correct program to output the calculation results of "50+100" to the console.**

a) console.log("50" + "100");

b) console.log(50" + "100);

c) console.log(50 + 100);

d) console.log("50 + 100");

2. **Select the correct symbol for the division used in the program.**

a) *

b) /

c) #

d) ÷

3. **Let the computer calculate "19+40" and output the calculation results to the console.**

....

4. **Let the computer calculate "7-2" and output the calculation results to the console.**

....

5. **Let the computer calculate "15 divided by 5" and output the calculation results to the console.**

....

6. **Select the correct program to output "Today is the 17th.7 days ago was the 10th." to the console.**

a) console.log("Today is the 17th.7 days ago was the + 17 - 7 + th.");

b) console.log("Today is the 17th.7 days ago was the " + "17 - 7" + "th.");

c) console.log("Today is the 17th.7 days ago was the " + (17 - 7) + "th.");

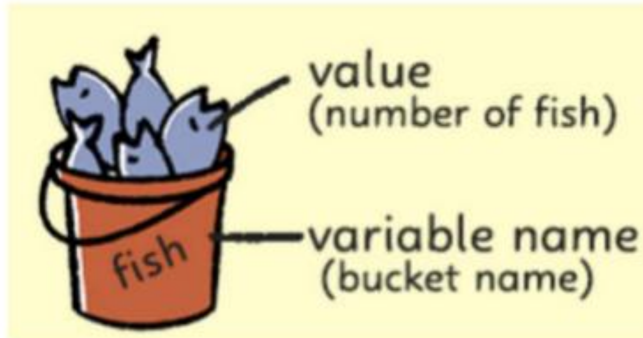
d) console.log("Today is the 17th.7 days ago was the (17 - 7)th.");

7. **Let the computer calculate "2*5" and output "The answer is 10." to the console.**

....

Chapter 3
Variables

- **Variable**: It is like a box for storing a value.
- Each variable has a value.
- The value of a variable can be changed.



Step 1: Declare `let item1;` → **Step 2: Assign** `item1 = 25;` → **Step 3: Use** `console.log(item1);`

EX Declare the variable fish, assign 0, assign fish+8, assign fish+12, and create a program to output the value of the variable to the console.

Solution:

Program:

```
let fish;
fish = 0;
fish = fish + 8;
fish = fish + 12;
console.log(fish);
```

Output:

Sample: Output

```
20
```

Revision

1. Create a variable number, Assign 100 to the variable number. then output the value of number to the console.

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2. In the previous program, (1) Assign the value of the variable number plus 10 (2) Output the value of variable number to the console

   ……………………………………………………………………………………

   ……………………………………………………………………………………

3. What will be the output of the following code ?

```
let a = 5;
a = a + 3;
console.log("The result is " + a);
```

   ……………………………………………………………………………………

Chapter 4

Mid-Term Exam

1. Select the correct program to output "Hello" to the console.

   a) console.log("Hello"); b) consolelog("Hello")

   c) console.logHello; d) log.console("Hello");

2. Output "Hello" to the console.

   ....

3. Create your program as follows:

   -Change "Leave this text." to a comment.

   -Execute the program and it will output "Make the first line a comment."

   JavaScript
   ```
   Leave this text.
   console.log("Make the first line a comment.");
   ```

   ....

   ....

4. Select the correct program that will let the computer calculate "5+7" and output the calculation results to the console.

   a) console.log(5" + "7); b)console.log("5 + 7");

   c) console.log("5" + "7"); d) console.log(5 + 7);

5. Select the correct symbol for multiplication used in the program.

   a) / b) + c) - d) *

6. Let the computer calculate "3*5" and output "The answer is 15." to the console.

   ....

   ....

7. Select the correct declaration for the variable number.

   a) let = number; b) number let;

   c) number; d) let number;

8. Create your program as follows

   (1) Declare the variable num.

   (2) Assign 5 to num

   (3) Output the value of num to the console

   ....

   ....

Chapter 5
If statement

Case1: Test 1 condition

If (condition) {processing : This code is executed if condition is true}

EX Declare the variable science, Assign "100" to it, and if the variable science is "100", Output "Excellent".

Solution

JavaScript

Run Code ▶

```javascript
let science = 100;
if (science == 100){console.log("Excellent");}
```

Console

```
Excellent
```

NOTES:

1. We use "=" : to assign value to a variable
2. We use "==": to compare two values.

You can use conditional expression without an if statement, The result will be (True or False)

JavaScript

Value assigned to the variable science

```javascript
let science = 100;
console.log (science == 100);
```

Compare if the values are the same

Execute...

Console

```
true
```

Revision

1. Create your program as follows.
   (1) Declare the variable num and assign 5 to it.
   (2) Output the value of variable num to the console.
   However, the entire program should be written on two lines (the declaration and initial value assignment should be written on one line).
   ……………………………………………………………………………………………………
   ……………………………………………………………………………………………………
2. Select a program that says, "If ~, then ________."
   a) (conditional) if {processing}
   b) (conditional){processing} if
   c) if {processing}(conditional)
   d) if (conditional){processing}
3. Select the conditional expression "the value of the variable num is equal to 50".
   a) 50 >= num
   b) num = 50
   c) num == 50
   d) 50 = num
4. Create your program as follows
   (1) Declare the variable num and assign 100.
   (2) If variable num is equal to 100, output "Correct".
5. Please select the output result of the following program.

```
let num = 200;
console.log(num == 100);
```

   a) true
   b) num
   c) 100
   d) false

Comprehensive revision on chapters (1-5)

1. Which line has an error?

```
console.log(see you tomorrow);
console.log("see you tomorrow");
```

2. Write a JavaScript program to calculate 19 + 40 and output the result.

   ………

   ………

3. Select the correct way to declare a variable:

   a) let = number;   b) number let;

   c) number;   d) let number;

4. Write a JavaScript program that calculates the total cost of three shopping items and gives a message based on the total amount.

   Instructions:

   1. Add the following comment at the top of your program :" *This program calculates total shopping cost"*
   2. Create three variables named item1, item2, and item3.
   3. Assign any numeric values to them (25, 15, 10).
   4. Calculate the total cost and store it in a variable named total.
   5. Output the price of each item and the total cost.
   6. Calculate and output the average price per item.
   7. Use an if statement:
      - If total is **50 or more**, output "Get a discount!".
      - Otherwise, output "Add more items to get a discount!".

   ………

   ………

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Chapter 6
if-else statement

Case2

- It is used when you want to execute processing when the conditional expression is **true** or **false**, respectively
- If (condition) { A }
  else { B }

Comparison operators *Not all

1	A==B	A is the same as to B
2	A>B	A is greater than B
3	A<B	A is less than B
4	A>=B	A is greater than or equal to B
5	A<=B	A is less than or equal to B

EX Declare variable Math, Assign 80 to it.
Output "Excellent" if the variable Math is greater than or equal 90.
If not, the program outputs "You need to study hard ".

Solution

JavaScript | Run Code ▶

```javascript
let math = 80;
if (math >= 90) {console.log("Excellent");}
else {console.log("You need to study hard");}
```

Console

```
You need to study hard
```

Review

1. Select a program that says if ~ then do A, otherwise do B.
   a) if (conditional){A}
      else{B}
   b) if (conditional) else{B}
      {A}
   c) else (conditional){A}
      if {B}
   d) else{B}
      if (conditional){A}
2. Create your program as follows.
   (1) Declare the variable num and assign 5 to it.
   (2) If variable num is the same as 10, output "Win".
   (3) Otherwise, output "Lose".
   ....
   ....
   ....
   ....
3. Select a program with the conditional expression "the value of the variable num is less than or equal to 80".
   a) num < 80 b) num == 80
   c) num <= 80 d) num > 80
4. Select a program with the condition "the value of the variable num is greater than 30".
   a) num < 30 b) num == 30
   c) num <= 30 d) num > 30
5. Create your program as follows
   (1) If the variable score is greater than or equal to 60, "Pass" is output; otherwise, "Fail" is output.
   Please do not delete the program that has already been written and input the conditional expression part.
   ....
   ....
   ....
   ....

Chapter 7
else if statements

Case3: Test <u>multiple</u> conditions

If (condition 1) { A }
else if (condition 1) { B }
else if (condition 2) { C }
else if (condition 3) { D }

Create your program as follows
(1) Declare the variable "egg" and assign 5 to it.
(2) If the variable egg is 0, output "No stock".
(3) If not, and the variable egg is less than or equal to 3, output "Low stock".
(4) If not, output "In stock".

Solution

JavaScript

```javascript
let egg = 5;
if (egg == 0){console.log("No stock");}
else if (egg <= 3){console.log("Low stock");}
else {console.log("In stock");}
```

Console

```
In stock
```

Note: A program is executed sequentially, starting from the top.

Review

1. Choose a program that says, "If ~, do A; if not, do B."

a) if (conditional){A} else if {B}	b) if (conditional){A}{B} else if (conditional)
c) if (conditional){A} else if (conditional){B}	d) if (conditional) (conditional) else if {A}{B}

2. select the output result of the following program.

```
let score = 62;
if (score == 100){console.log("Perfect score");}
else if (score >= 80){console.log("So close");}
else if (score >= 60){console.log("Pass");}
```

a) So close　　b) Perfect score

b) Pass　　d) No output

3. Create your program as follows

(1) Declare the variable "egg" and assign 5 to it.

(2) If the variable egg is 0, output "No stock".

(3) If not, and the variable egg is less than or equal to 3, output "Low stock".

(4) If not, output "In stock".

Chapter 8
Revision (Mid-term)

1. Create your program as follows
   (1) Declare the variable num and assign 7.
   (2) If variable num is equal to 7, output "Correct".

   ……………………………………………………………………………………………………

   ……………………………………………………………………………………………………

2. Please select the output result of the following program.

```
let num = 10;
console.log(num == 10);
```

a) 10	b) true
c) false	d) 100

3. Select a program with the conditional expression "the value of the variable num is less than 20".

a) num >= 20	b) num <= 20
c) num == 20	d) num < 20

4. Create your program as follows.
   (1) Declare the variable num and assign 100 to it.
   (2) If variable num is equal to 100, output "correct". (3) If not, "miss" is output.

   ……………………………………………………………………………………………………

   ……………………………………………………………………………………………………

5. Please select the output result of the following program.

```
let score = 92;
if (score == 100){console.log("Perfect score");}
else if (score >= 80){console.log("So close");}
else if (score >= 60){console.log("Pass");}
```

a) So close	b) Perfect score
c) No output	d) Pass

Chapter 9
Logical operator

<table>
<tr><th>Symbol</th><th>Name</th><th>meaning</th><th>Example</th></tr>
<tr><td>&&</td><td>AND</td><td>Both conditions should be true</td><td>if (age > 18 && score >= 50){ }</td></tr>
<tr><td>||</td><td>OR</td><td>Only one condition should be true</td><td>if (age >= 18 || age < 18){ }</td></tr>
</table>

&: ampersand |: Vertical bar

Create your program as follows.

(1) Declare the variable **numberOfPeople** and assign 2.
Declare the variable **fee** and assign 3500.
If the variable **numberOfPeople** is greater than or equal to 2 and the variable **fee** is greater than or equal to 3000, "Free mug" is output to the console.

Create your program as follows.

(1) Declare the variable **age** and assign 15 to it.
(2) If the variable **age** is less than or equal to 12 or greater than or equal to 65, "30% off" is output to the console.

Review

1) Select a program that means "A and B"

a) A && B	b) A // B
c) A & B	d) A || B

2) Select the program that means "A or B".

a) A && B	b) A // B
c) A & B	d) A || B

3) Please select the output result when executing the following program.

```
let age = 20;
if (age <= 18 || age >= 60){console.log("20% OFF");}
else{console.log("Regular Price");}
```

a) Error	b) No output
c) Regular Price	d) 20% OFF

4) **Please create a program as follows.**

If variable a is equal to 5 or b is equal to 10, output "Condition met".

```
let a = 5;
let b = 10;
if (                    ){console.log("Condition met");}
```

5) Please create a program as follows.

If variable a is greater than or equal to 2 and b is greater than or equal to 3, output "Condition met.

```
let a = 4;
let b = 3;
if (                    ){console.log("Condition met");}
```

Chapter 10
Iterative operation
For statement (Loops)

for (start; condition; update){ }

JavaScript

```
for (let count = 0; count < 10; count = count + 1 ) {
  console.log("silk");
}
```

Define variables — Conditional expressions — Update variables

Process

JavaScript

Necessary Necessary Not necessary

```
for (let count = 0;count < 10;count = count + 1;) {
  console.log("silk");
}
```

";" (semicolon) means "a separator for the program," so it needs to be placed between parts of the program.

NOTES: **1- count = count + 1** can be written as **count++**

2- in the programming world, in for statement ( ) , the variable is often named **i**

EX Create your program as follows.

(1) Output "campus" 10 times to the console.
The entire program should be 3 lines.
Variable name should be **count**.

NOTE: **1-** Break a line after the left curly bracket " **{** " and before the right curly bracket "}".

2- Use <u>indentation</u> (space at line start) to make code easier to read

EX

```
if (numberOfPeople >= 2 && fee >= 3000) {
  console.log("Free mug");
}
```

Revision

1) Select the first thing you would write in a program that represents "outputting the word "apple" to the console 10 times over.

a) let	b) for
c) num	d) if

2) Select the correct program to add 1 to variable i.

a) i+	b) i++
c) 1+1	d) i++1

3) Please select the correct answer for the program that repeats the specified process 10 times.

a) for (i<10; let i=0; i++) {
Specified process
}
	b) for {Specified process} (
	let i=0; i<10; i++
	)

c) for (let i=0; i<10; i++) {
Specified process
}
	d)
	for (let i=0; i++; i<10) {
	Specified process
	}

4) Create your program less than or equal to
Output "Hello" to the console 5 times.
*Variable name is i and assign 0 at the beginning.
*Use the increment operator.

```
for (   ) {
  console.log("Hello");
}
```

………

………

5) Create your program less than or equal to
Output "Hello" to the console 7 times.
*Variable name is i and assign 0 at the beginning.
Use the increment operator.

………

………